המכון למחקרי נפט וגיאופיסיקה
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

MAGNETOTELLURIC SURVEY IN THE TAMAR AREA

By

A. Klang, Y. Rotstein, S. Goldberg

Prepared for

OFFICE OF THE CHIEF SCIENTIST
MINISTRY OF INDUSTRY, TRADE AND TOURISM

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2. Apparent Resistivity and Phase Data of the Magnetotelluric Soundings.

ABSTRACT

In an attempt to discover the significance of a low apparent resistivity anomaly detected in roving dipole measurements, four magnetotelluric soundings were measured in the Tamar area. The results of this survey indicate anisotropic apparent resistivities in all the measured soundings.

Only a qualitative interpretation could be carried out. The possibility of a low resistivity anomaly in the area of site TR-4, added to the effect of a two dimensional structure, is considered.

It is recommended that one or two wells be drilled at 1.5 Km west of Tamar 9 well to a depth of 150-200 m and the temperature gradient be measured in them.

1. INTRODUCTION

This report describes results of four magnetotelluric soundings carried out by the Institute for Petroleum Research and Geophysics in the Tamar area (Fig. 1), from October, 1977 to March, 1978. The survey was performed as part of the geothermal program in Israel and sponsored by the Chief Scientist, Ministry of Industry, Trade and Tourism. Its aim was to discover the significance of a low resistivity anomaly detected by roving dipole measurements (Klang and Levanon, 1978) in the Tamar area.

The magnetotelluric sites were not situated in the central part of the anomaly near Tamar 9 well. Due to the inaccessibility of this area, the soundings were measured at approximately 1.5 Km east (TR-1) and 3.0 Km west (TR 2-4) of the well.

Results of the different soundings and special interpretation problems, arising from the complex structure in which the soundings were located, will be discussed. Conclusions and recommendations concerning the surveyed area conclude this report.

2. GEOLOGY

The surveyed area (Fig. 1) is located in the vicinity of the Tamar wells, in the Northern Negev southwest of the Dead Sea. The region consists of two very different geomorphological units separated from each other by a north-south trending fault escarpment which lies along the western edge of the Dead Sea - Arava depression. East of this escarpment, constituting a prominent morphotectonic line, extends the Rift Valley while to the west of it lies a mountainous area.

The Dead Sea Rift Valley is part of the Syrian-African Rift system and was formed in late Pliocene to early Pleistocene times (Neev and Emery, 1967).

The mountainous area west of the fault escarpment is composed of two major morphotectonic units, extending from SSW to NNE. Of interest in this work is the eastern unit, namely Hatsera (Hamaktesh Haqatan) - Mount Hamarmar anticline, located at the western edge of the surveyed area (Bentor and Vroman, 1960). The eastern flank of this structure is rather steep; a typical feature of all the anticlinal folds and flexures in the Negev.

The general stratigraphy in the vicinity of the Dead Sea has been investigated on the basis of outcrops by several geologists and the results have been compiled by Bentor et. al. (1959) and Bentor and Vroman (1960). Additional data was obtained from wells which were drilled in the area. The strata are grouped into pre-graben and post-graben formations.

Strata deposited before the main faulting and the subsidence phase of the graben are widely exposed on both sides of the graben. They consist mainly of Precambrian igneous rocks, Nubian (Paleozoic to Lower Cretaceous) sandstones, Cretaceous to Lower Tertiary dolomites, limestones and marls, Neogene argilleous shales, sandstones and conglomerates.

The post-graben formations constitute the Dead Sea Group which includes a thick salt sequence (Late Pliocene to Middle Pleistocene) interfingering with a gravelly and marly sequence. This sedimentation is followed by whitish chalk, marl and gypsum, oolitic and sandy chalk and Tufa (Lisan Formation - Late Pleistocene). The following post-Lisan sediments are characterised by layers of coarsely crystalline rock salt interfingering with clays, silts, sands and gravels.

A number of water wells were drilled in the Tamar area near the fault. Out of these, Tamar 9 well, located on the up-side of this normal fault and drilled to a depth of 450 m in the Lower Cretaceous, indicated an average geothermal gradient of 135.1 ± 30.24 $^{o}C/Km$ and a very high heat flow of 12.24 ± 4.74 $\mu cal/cm^2 sec$ (Eckstein, 1976). Unfortunately, temperature measurements could only be performed

to a depth of 120 m. Eckstein (1976) explains this high heat flow anomaly as the effect of a shallow warm water reservoir, created by the ascending water along the rift fault from a great depth. According to this explanation, no high temperature reservoir can be expected to exist at depth and the region is of no geothermal interest.

3. FIELD PROCEDURE AND INSTRUMENTATION

Four sites were measured in this survey. Site TR-1 is located in the Dead Sea - Arava depression, about 1.5 Km southeast of Tamar 9 well. Sites TR-2 to TR-4 were placed about 3 Km west of Tamar 9 well in a mountainous area which restricted access to more favourable locations (Fig. 1).

A five-channel digital magnetotelluric system, manufactured by Geotronics Inc., was used in this survey. Induction coils were employed to record changes in the magnetic field. $Cd\text{-}CdCl_2$ non-polarized and Pb electrodes were used to detect the telluric current.

The sites were recorded in the frequency range of about 50 to 10^{-3} Hertz and the recording over this wide (over five decades frequency) range was divided into six bands.

4. DATA ANALYSIS AND INTERPRETATION

The field tapes were analysed following the procedure described by Sims et. al., (1971). Tensor apparent resistivity values were rotated to their principal directions; corresponding to the geoelectric strike and perpendicular to it, namely TE and TM. Phases for each of the apparent resistivity components were also computed and used in the analysis. Data with phasor coherencies of 0.9 and better were used in the interpretation except for occasional instances where data

with coherencies of 0.8-0.9 were judiciously added. These low coherencies were added only to supplement existing higher coherency values at the same frequencies, when they did not generate any irregular complexity to the data.

In general, the interpretation in magnetotellurics is based on models consisting of horizontal homogeneous isotropic layers. In this case the resistivity is changing only with depth and no lateral variations are considered.

In such cases, the interpretation is based on a one dimensional inversion of the TE data which yields a resistivity cross-section. The resistivity cross-section, obtained by an automatic inversion process, approximates the true resistivity section under the site. It can be extended to cases where the subsurface structure is two dimensional with mild dips.

In the present survey, however, the sites are located in complex geologic situations. Site TR-1 was located in the Rift Valley at about 1.5 Km from the western fault of the graben. In this situation, the resistivity is changing not only with depth, but also significant lateral changes of resistivity must be taken into account when interpreting the measured data of this site.

Sites TR-2 to TR-4 are located near the steep eastern flank of a major anticlinal structure. Under these conditions, lateral variations of the resistivity should also be expected. It is clear that in the present survey, one must deal with the effects of complex structures whose interpretations are based on two - dimensional and even three-dimensional models.

As shown by Patrick (1969), two-dimensional models are not amenable to any inversion fitting procedures. Not only are they more expensive to compute but the multitude of possible parameters makes the problem intractable. Some specific models were resolved using finite difference techniques (Patrick, 1969; Wright, 1970).

Based on these considerations, this interpretation conducted in a complex geologic environment will be limited to some qualitative considerations and an effort to compare the present results with computed two-dimensional models will be made.

5. RESULTS AND DISCUSSION

Apparent resistivity and phase data of four soundings are presented in Figure 2. In sounding TR-1, the two distinct apparent resistivities (TE and TM) are clearly separated from each other and indicate the two dimensional effect produced by the strong lateral change of the resistivity, due to the fault. As previously discussed, it is impossible to accurately define the true resistivity section under the site at these conditions. Qualitative analysis of the resistivity variation with depth is presented.

At this site, near surface resistivity values in the range of 15-30 ohm-m are indicated by the apparent resistivity values for periods less than 0.1 sec. The resistivity appears to strongly decrease (possibly to less than 6 ohm-m) with depth. The thickness of this low resistivity layer cannot be clearly defined. This second layer overlies a third one with high resistivity, indicated by increasing apparent resistivities for the long periods (10^2-10^3 seconds).

This resistivity section is in accordance with the model found by Paparo et. al. (1977) in their reinterpretation of Schlumberger and Wenner soundings of the Northern Arava region. For a sounding located about 2 Km eastward of the MT site TR-1, they found a near surface layer with a resistivity of 30 ohm-m; followed by a low resistivity layer (7.5-0.48 ohm-m) section to a depth of about 400 m. A high resistivity layer appears below this depth (Paparo et. al., 1978; Fig. 90).

The near surface resistivity value in the range of 15-30 ohm-m, as interpreted from the magnetotelluric data, also correlates with data obtained from roving dipole mapping with the sources located in the vicinity of Tamar 9 well (Klang and Levanon, 1978; Fig. 21-22).

The apparent resistivity values measured at dipole sites located in the vicinity of the sources are mainly affected by the resistivities of near surface layers. Therefore, good correlation between the dipole data and the magnetotelluric first resistivity layer can be expected.

MT sites TR-2 to TR-4 are located in the mountains area about 3 Km west of the graben fault. For each of the sites, two distinct TE and TM apparent resistivity curves are observed, indicating anisotropic apparent resistivities. For periods lower than 10 sec., the difference between TE and TM apparent resistivities is much larger at site TR-2 than at TR-3. For longer periods, TE and TM apparent resistivities converge and at very long periods (higher than 100 sec.), the TM apparent resistivities are higher than the TE ones at both soundings. Sounding TR-4 indicates the strongest anisotropy of the apparent resistivity. TE and TM data are completely separated over the whole frequency range. TE apparent resistivity values at TR-4 are one order of magnitude less than those observed in soundings TR-2 and TR-3. They are also lower than the TM apparent resistivity of the same sounding.

The results presented here indicate a non-homogeneous structure with the sounding TR-4 being located nearest to it. The following two factors, contributing to the non-homogeneity of the structure, must be considered when studying the anisotropy:

1) The graben - The lateral effect of the low resistivity layers which appear in the graben.

2) The strong dips of structures - The steep eastern flank of Hatsera - Mount Hamarmar anticline.

Of the two factors, the first should equally affect all soundings made at sites situated approximately the same distance from the graben. The second factor's contribution could not be fully evaluated. The sites are located at about the same distance from the steep flank of the Hatsera structure. From the south, they approach the local structure of Mount Tsurim dome (Bentor and Vroman, 1960) which is situated about 1.5 Km east of the sites.

Based on these considerations, it becomes clear that the strong changes in apparent resistivity appearing at each site cannot be explained by the lateral change of resistivities in the graben. Rather, it seems that at sounding TR-4, a lower resistivity anomaly has been added to the described structural factors. This may better explain the much lower TE apparent resistivities as compared with those in soundings TR-2 and TR-3.

A possible indication as to the validity of such an interpretation can be found in the work of Lee et. al. (1978) which includes magnetotelluric apparent resistivity pseudosections over some two dimensional models. One of these models, model 5A, includes a thin low resistivity dyke intruding a high resistivity layer; the TE apparent resistivity values decrease over the low resistivity anomaly while the TM apparent resistivities remain nearly constant.

At the present stage of magnetotellurics, no direct quantitative interpretation of the anisotropic apparent resistivities can be achieved without cumbersome and expensive computations and the number of two dimensional models used in the above-mentioned problem are not sufficient to cover the real geologic situations encountered in field work.

In conclusion, it seems to us that no definite evaluation can be given at this

stage as to the validity of a low resistivity anomaly in the area of sounding TR-4. If it exists, its effect is additional to the effect of the non-homogeneous structure known to be present in the area.

The low apparent resistivity anomaly detected by the roving dipole measurements (Klang and Levanon, 1978) is also not clearly defined in the Tamar wells area. This anomaly is detected by roving dipoles located outside the graben (high resistivity medium) with the sources located inside the graben and near the graben fault. In this case, which can be considered as a single vertical contact structure, the computed dipole resistivity effects show that the apparent resistivity on the resistive side of the contact will be less than the true value (Furgerson and Keller, 1974). The exact amount of reduction cannot be calculated and the anomalous zones mapped for the two sources (Klang and Levanon, 1977; Fig. 21-22) may have, in reality, higher values.

In addition, this low apparent resistivity anomaly is not clearly delineated. On the map of Source 19, the anomaly is based on only one measurement and is not consistent in the two maps, having different shapes.

Tamar 9 well which is located in the area indicated a high temperature gradient ($135 \pm 30^{o}C/Km$) at a shallow depth (up to 120 m). Eckstein (1976) explains this gradient by the presence of ascending hot water along the border fault, contributing to convective heat flow in the area.

As already shown, magnetotellurics did not help clarify the significance of this low resistivity zone. Consequently, no conclusive response can be given as to the validity of the apparent low resistivity anomaly detected by dipole mapping.

6. CONCLUSIONS AND RECOMMENDATIONS

The present survey, aimed at finding the significance of the low apparent

resistivity anomaly detected by roving dipole measurements in the Tamar area, could not further clarify the problem. The magnetotelluric survey was carried out in a complex geological area and anisotropic apparent resistivities were obtained at the measured soundings. Only a qualitative interpretation could be achieved and the possible presence of a low resistivity anomaly in the area of site TR-4, added to that of the two dimensional structure, was discussed.

At the present stage of magnetotellurics, no direct interpretation could be carried out to confirm this supposition and no computed two dimensional model could be fitted entirely to the complex geological environment in which this survey was performed. The high thermal gradient found in Tamar 9 well may be caused by the presence of ascending hot water along the border fault, but this in itself cannot be considered as strong evidence to the presence of a thermal source at depth. No further magnetotelluric or electrical surveys are proposed for this complex geological area.

The drilling of one or two wells (1.5 Km west of Tamar 9 well) to a depth of 150-200 m and the measurement of their temperature gradients is recommended as a possible method for finding the geothermal significance of the low resistivity anomaly.

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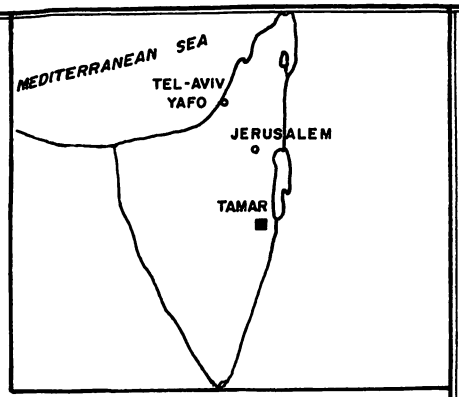
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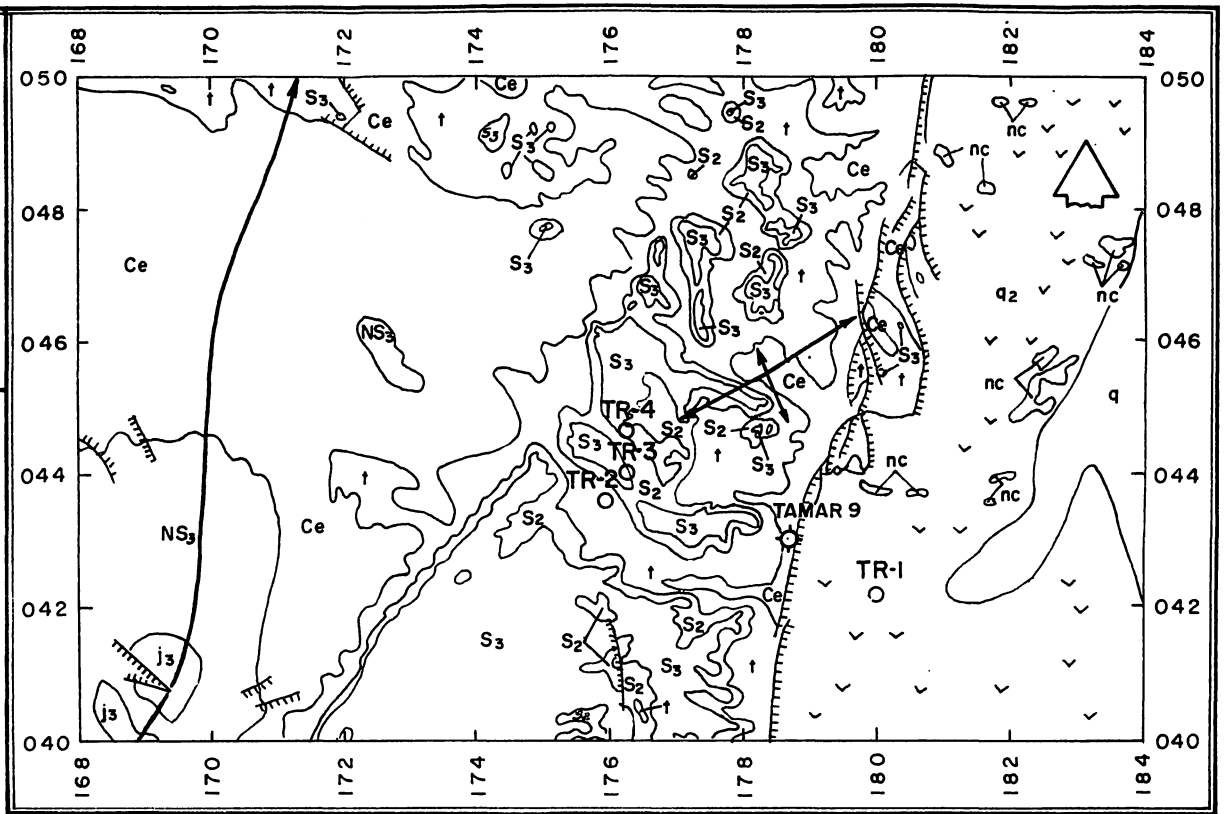
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LEGEND

- Quaternary — q
- Quaternary (Lisan Marl) — q2
- Neogene (Hatseva Series) — nc
- Campanian — S3
- Santonian — S2
- Turonian — t
- Cenomanian — Ce
- Lower Cretaceous — NS3
- Jurassic (Malm) — j3
- Magnetotelluric Sounding — TR-1
- Well
- Dip
- Fault
- Hatsera Anticline
- Mt. Tsurim dome



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ISRAEL

LOCATION and GEOLOGICAL MAP

MAGNETOTELLURIC SURVEY

TAMAR AREA

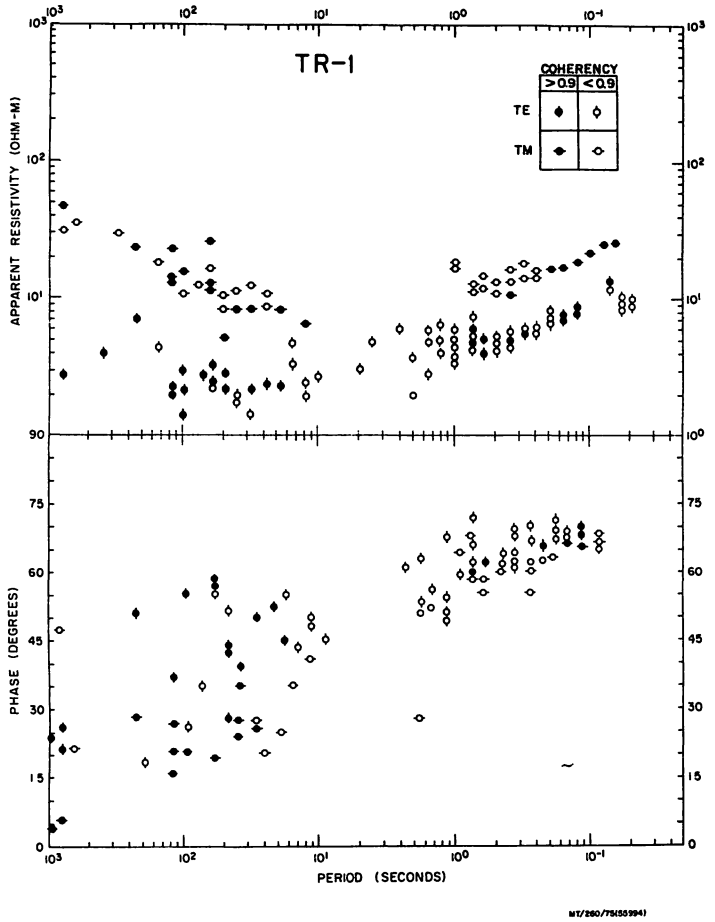
SCALE
0 2 4 km.

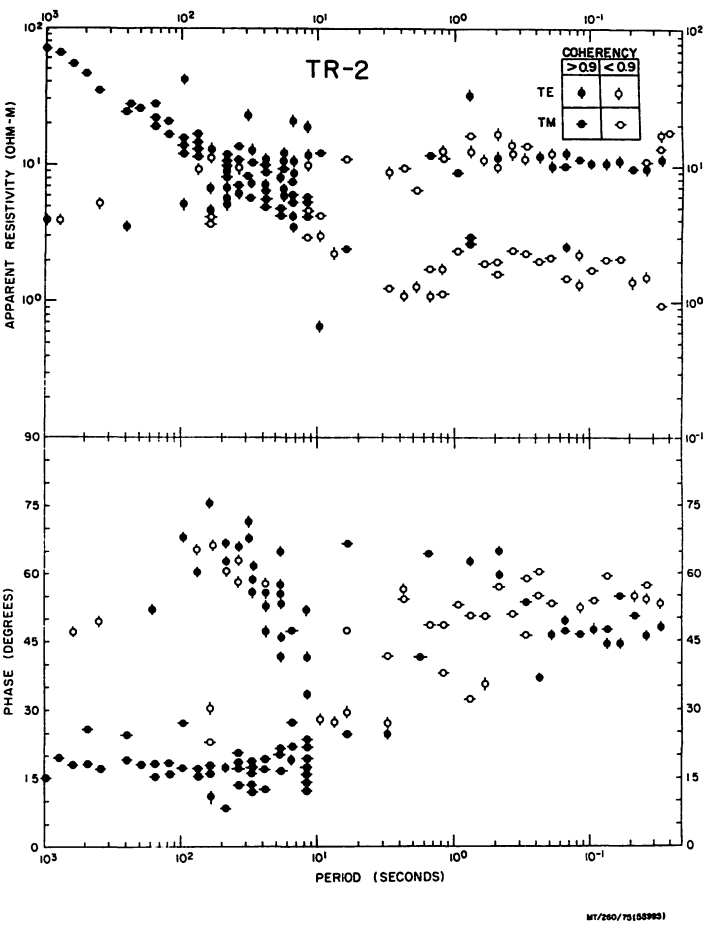
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MOUNT SEDOM, BENTOR and VROMAN

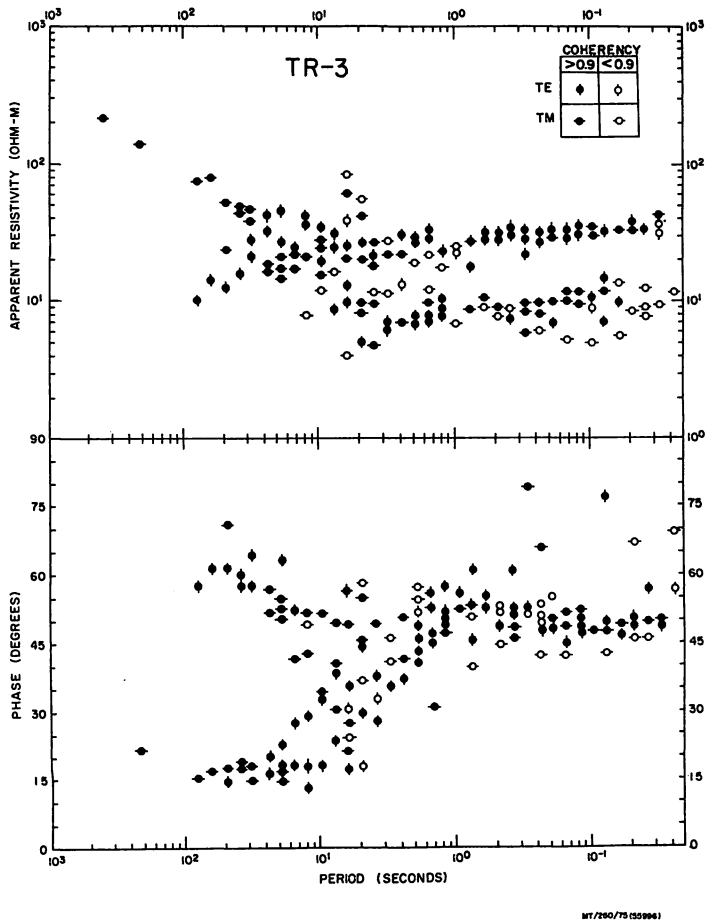
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Figure 1


TR-1
COHERENCY
>0.9 <0.9
TE
TM
APPARENT RESISTIVITY (OHM-M)
PHASE (DEGREES)
PERIOD (SECONDS)


TR-2
COHERENCY
>0.9 <0.9
TE
TM
APPARENT RESISTIVITY (OHM-M)
PHASE (DEGREES)
PERIOD (SECONDS)


TR-3
COHERENCY
>0.9 <0.9
TE
TM
APPARENT RESISTIVITY (OHM-M)
PHASE (DEGREES)
PERIOD (SECONDS)

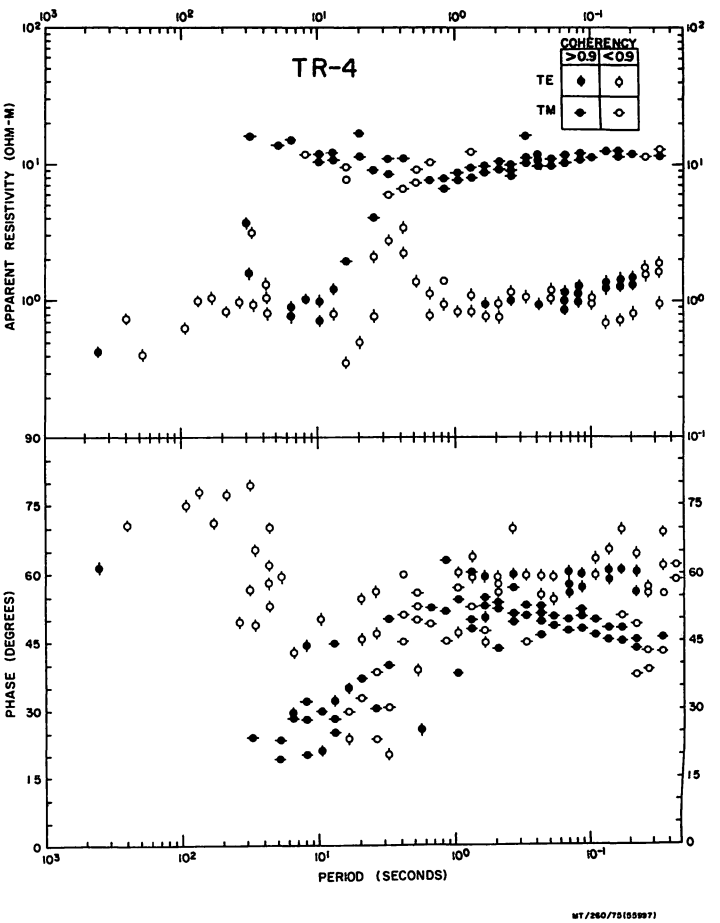

TR-4
COHERENCY
>0.9 <0.9
TE
TM
APPARENT RESISTIVITY (OHM-M)
PHASE (DEGREES)
PERIOD (SECONDS)

Figure 2